

LaTeX Typesetting & Technical Formatting

A polished portfolio sample for mathematical, academic, and technical documents.

PORTFOLIO
Self-created sample

- EQUATIONS
- TABLES
- TIKZ / Pgfplots
- ACADEMIC LAYOUT
- REFERENCES

What this sample demonstrates

CLIENT-SERVICE FORMAT

Accurate mathematical notation, structured technical layouts, vector-style figures, professional tables, consistent typography, and maintainable document structure - presented as a clear, client-facing typesetting workflow.

Core Formatting Services

01 MATH Mathematical Typesetting Equations, matrices, aligned derivations, symbols, operators, integrals, and structured notation.	02 LAYOUT Academic Document Formatting Reports, assignments, project documents, thesis-style layouts, headings, captions, and page structure.
03 FIGURES Technical Figures TikZ diagrams, mathematical plots, annotations, axes, labels, legends, and Pgfplots visualizations.	04 DATA Tables & References Publication-style tables, captions, cross-references, bibliographic structure, and consistent presentation.

At a Glance

- Professional mathematical notation
- Clean visual hierarchy
- Editable LaTeX source
- Vector-based technical graphics
- Structured tables and captions
- Cross-referenced document elements

Ideal for

- Research notes and technical reports
- Mathematics assignments and solutions
- Project reports and academic submissions
- Thesis / dissertation formatting
- Lecture notes and handouts
- Data and computational reports

Typical delivery

Compiled PDF + editable .tex source when required, with a structured and maintainable document.

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Technical Showcase

Selected examples of mathematical typesetting, technical graphics, tables, and document structure.

01 / MATHEMATICAL TYPESETTING

Singular Value Decomposition

$$A = U\Sigma V^T, \quad A \in \mathbb{R}^{m \times n}.$$

Block Matrix / State-Space Form

$$\mathbf{M} = \begin{bmatrix} A & B \\ C & D \end{bmatrix}, \quad x_{k+1} = Ax_k + Bu_k.$$

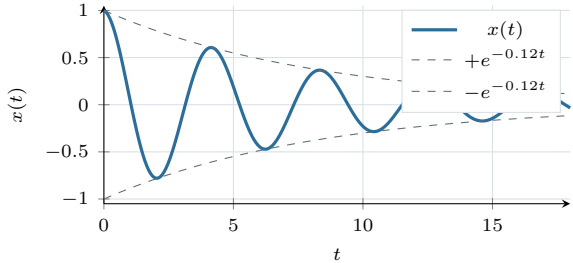
Aligned Derivation

$$I = \int_{-\infty}^{\infty} e^{-x^2} dx,$$
$$I^2 = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)} dx dy.$$

Aligned environments, matrices, symbols, automatic spacing, and clean equation hierarchy.

02 / TECHNICAL GRAPHICS

Illustrative Damped Oscillation



Vector-style mathematical visualization generated directly in $\text{\LaTeX}$ using TikZ/Pgfplots.
Consistent fonts, mathematical labels, axes, legends, and reusable figure code.

Structured Tables and Presentation

Method	Order	Step Size	Iterations
Explicit Euler	1	0.010	2000
Midpoint Method	2	0.010	2000
Classical RK4	4	0.010	2000

Illustrative values shown solely to demonstrate professional table formatting; not reported experimental results.

03 / DOCUMENT STRUCTURE

- Automatic equation, table, and figure references
- Consistent heading hierarchy
- Caption and spacing control
- Maintainable section structure
- Clean PDF compilation

04 / CLIENT WORKFLOW

- Review source material and requirements
- Structure content and formatting rules
- Typeset and refine the document
- Quality-check layout and references
- Deliver PDF and editable source

